

ARYAA VIJAY

Aerospace Engineering | Hardware Integration, Controls & CFD

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Portfolio: aryaa207.github.io

Melbourne, FL | U.S. citizen | Available May 2027: full-time or internship

EDUCATION

Florida Institute of Technology - B.S. Aerospace Engineering

Expected May 2027 | **GPA 3.8 / 4.0** | Dean's List: Fall 2024, Fall 2025

Coursework: Control Systems, Aerodynamics, Compressible Flow, Fluid Mechanics, Thermodynamics, Aerospace Structural Design, Materials Science, Computational Techniques.

TECHNICAL PROJECTS

S.A.R.D. Quadcopter & Vision Payload

2025 - 2026

- Independently integrated a quadcopter platform, flight controller, sensor hardware, and Raspberry Pi 4B vision software using Python and OpenCV.
- Designed and 3D-printed a two-axis camera mount in Onshape with micro-servo integration; assembled and demonstrated the physical hardware and camera interface.
- Built an LSM9DS1 acquisition dashboard with startup gyro-bias sampling, roll/pitch estimation, and WebSocket streaming to the browser.

3D Guidance Simulation - PN / APN

April 2026

- Primary contributor to MATLAB simulation implementation and analysis in a three-person presentation team; compared True and Augmented Proportional Navigation.
- Developed a randomized evaluation framework and analyzed trajectory and acceleration histories; final code and presentation document a 200-trial ensemble.

Mach 5 External Aerodynamics

January - February 2026

- Prepared geometry and a 3,389,998-cell tetrahedral mesh for ANSYS Fluent analysis using a density-based solver and SST k-omega turbulence model.
- Examined Mach and pressure contours to study shock structure and pressure distribution; documented model assumptions and convergence limitations.

Flying-Wing Glider - Project Manager

May 2025

- Led airfoil selection, Reynolds-number estimation, and glide-ratio prediction in a team of four; created NX CAD and drawings and coordinated fabrication and flight testing.

RESEARCH EXPERIENCE

Undergraduate Research Assistant - Florida Tech

Sep 2025 - Apr 2026

Functional Biomaterials Lab

- Prepared precision chemical solutions and material samples for microscopy and structural characterization; calibrated and operated 3D bioprinting equipment.
- Maintained laboratory records, recorded testing parameters, and compiled data summaries for senior research staff.

TECHNICAL TOOLS

Analysis: MATLAB, Python, ANSYS Fluent, Monte Carlo analysis, data acquisition

Design: NX Siemens, Onshape, KiCad, 3D printing

Embedded: Raspberry Pi, Arduino, IMU integration, OpenCV, Git, LabVIEW